Experiment 7: Cell Division and Cytogenetics

By Tyler Raymond Harvey

Important Questions to Consider

1)

Q) Which of the phases of mitosis did you see most and fewest examples of in your microscopic examination of onion root tips?

A) The most phase I saw in our sample is the interphase. The fewest phase I saw was the Anaphase.

2)

Q) What Conclusion can you draw about the relative length of each of the stages of mitosis from your examination of onion root tips?

A) That most of the time the cell is in the interphase stage, and the anaphase stage is the shortest phase.

3)

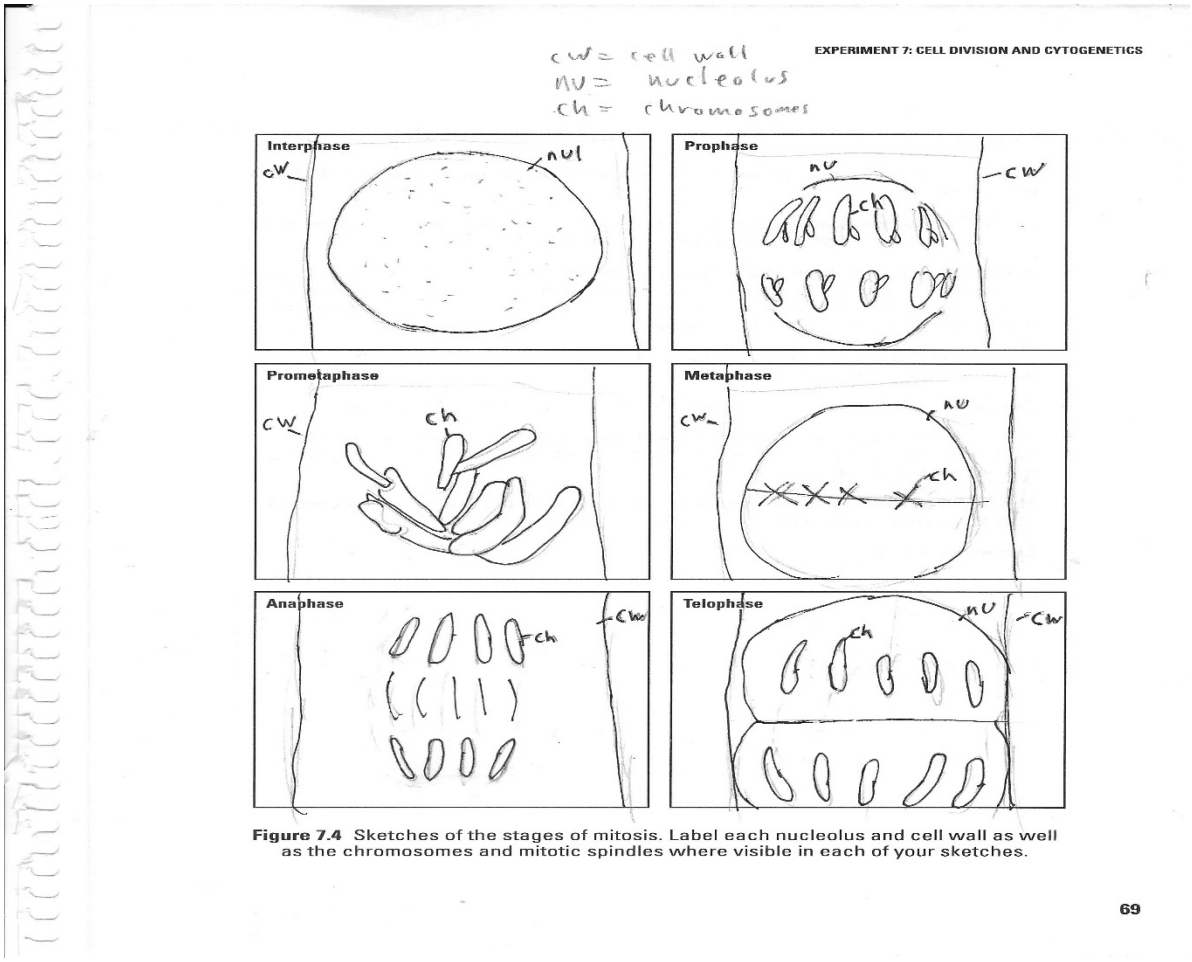
Q) What onion cells would you have to look at to examine meiosis as well as mitosis?

A) To look at both meiosis and mitosis inside onion cells, you must observe them within the meristematic region of the onion root tip.

4)

Q) Is the Chromosome you chose to sketch metacentric, submetacentric, or acrocentric?

A) It is acrocentric because it is chromosome pair #13 which is acrocentric.



cw = cell wall
nu = nucleolus
ch = chromosomes

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Figure 7.4 Sketches of the stages of mitosis. Label each nucleolus and cell wall as well as the chromosomes and mitotic spindles where visible in each of your sketches.

<table>
<tr><td>Case Study ID:
E</td><td>Patient Name:
Caroline</td><td>Age:
40</td><td>Sex:
Female</td></tr>
<tr><td colspan="2">Reasons for Referral:
May have a child with birth defects due to her age
-Her ultrasound she took revealed absent eyes, multiple fingers & toes with additional mutations of internal organs</td><td colspan="2">Chromosomal Finding
Total Number: 47
Type and number of sex chromosomes: XY)
X trisomy (Chromosome 13)</td></tr>
<tr><td colspan="4">Patient diagnosis:
Patav Syndrome Karyotype</td></tr>
<tr><td colspan="4">Implications of diagnosis (complications, life expectancy, etc.):
-Likely to result in spontaneous abortion.
-Baby will most likely die within the 1st week of life.
-Only 5% of babies will make it to year 1.
-Have severe cognitive disability and many physical abnormalities such as brain & spinal defect, cleft lips, palate, underdeveloped eyes, extra fingers, toes, and week muscle tone.</td></tr>
</table>

<table>
<tr><td>Case Study ID:
M</td><td>Patient Name:
Keri</td><td>Age:
Newborn</td><td>Sex:
Female</td></tr>
<tr><td colspan="2">Reason for referral:
Atypical characteristics such as disformed shape of ears which sets low on head, droopy eyelids, & minor webbing of the neck and between the fingers</td><td colspan="2">Chromosomal Findings:
Total Number: 45
Type and number of sex chromosome: X

X monosomy (Chromosome x)</td></tr>
<tr><td colspan="4">Patient diagnosis:
Turner Syndrome Karyotype</td></tr>
<tr><td colspan="4">Implications of diagnosis (complications, life expectancy, etc.):
Females with this illness only have one X chromosome.
-Clinical features can vary widely among individuals with signs including, short statue, low-set ears, a shorter/ webbed neck, heart defects, and kidney disorders.
-Will have premature ovary failure at some point during adolescence. Victims will not be able to enter puberty and begin menstruation, which will lead to infertility.</td></tr>
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